Smart Library Management System (BiblioTrack)



Project's Proposal

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1. Introduction

Libraries play an important role in academic learning by providing access to books, research materials, and other educational resources. However, many departmental libraries still rely on manual record keeping, which creates several problems such as difficulty in managing books, tracking issued items, calculating fines, and searching for required resources.

The Computer Science Department Library currently lacks a proper digital system to manage its operations efficiently. Students face delays when searching for books, checking availability, or receiving updates regarding issued books and fines.

To solve these issues, this project proposes the development of a Smart Library Management System (BiblioTrack). It is a web-based system designed to manage both physical books and digital resources. The system provides features such as smart search functionality, reservation system, automated fine management, digital PDF library, AI ChatBot assistance, and automated email notifications.

This modern solution will improve efficiency, provide instant access to information, and support students with smart library services.

2. Mission

The mission of this project is to transform the traditional departmental library into a modern smart digital library system that is efficient, user-friendly, and accessible to students anytime.

The system aims to:

- Replace the manual register system with a digital database.
- Provide quick and smart search functionality for books.
- Allow students to access digital learning resources (PDF books) online.
- Implement a reservation system for unavailable books.
- Provide AI ChatBot assistance for students to answer library-related queries.
- Automate fine calculation and E-Challan payment system.
- Send automated email notifications for account creation, book issue, reservation confirmation, and fine alerts.

- ❖ Maintain complete transaction records for transparency and reliability.

3. System Modules

3.1 Admin Module

The Admin Module allows the administrator to manage the entire library system.

Main functionalities include:

- ❖ Admin login authentication
- ❖ Dashboard overview of library activities
- ❖ Add, edit, and delete books
- ❖ Upload digital resources (PDF books)
- ❖ Manage registered students
- ❖ Manage new book requests
- ❖ Issue and return books
- ❖ Fine management and verification
- ❖ Verify E-Challan payments

3.2 *Student Module*

The Student Module allows students to interact with the library system through their accounts.

Main functionalities include:

- ❖ Student signup and login
- ❖ Profile management
- ❖ Search books by title, author, or category
- ❖ Check book availability
- ❖ Request new books

- View issued and returned books
- Download digital books (PDFs)
- Reserve unavailable books
- Generate clearance report

3.3 Digital Library Module

The system includes a Digital Library Section where the administrator can upload PDF books and digital learning resources.

Features include:

- Category-based organization of digital books
- Online viewing and downloading of PDFs
- 24/7 access to digital learning material
- Support for remote learning

3.4 Reservation System

The reservation system allows students to reserve books that are currently unavailable.

Key features:

- Students can reserve unavailable books
- Books are issued automatically when they become available
- Reservation follows a first-come, first-served queue system
- Students can cancel reservations anytime

3.5 AI ChatBot Assistant

An AI-based ChatBot is integrated into the system to assist students with library-related queries.

The ChatBot can help students with:

- Book recommendations

- Checking book availability
- Fine details
- Issued and reserved books information

If a query is unrelated to the library, the ChatBot responds with:

"Sorry, I can only answer library-related questions."

3.6 E-Challan System

The system includes an E-Challan payment system for handling library fines.

Features include:

- Automatic fine calculation for overdue books
- Generation of E-Challan with student details
- Payment through 1-Bill system
- Student confirms payment after submission
- Admin verifies and clears the fine

4. Techniques / Algorithms

The system follows a modular architecture and role-based access control between Admin and Student users.

4.1 Fine Calculation Algorithm

When a book is returned after the due date, the system calculates the fine automatically.

Formula:

Fine = Late Days × Fixed Rate

4.2 Reservation Algorithm

If a book is unavailable, the system adds the student to a reservation queue. When the book becomes available, the system automatically issues it to the next student in the queue following a first-come, first-served rule.

4.3 Email Notification System

The system sends automated email notifications using NodeMailer (SMTP service).

Emails are sent for:

- Account creation confirmation
- Book issue notification
- Reservation confirmation
- Book request submission
- Fine notification alerts

These automated emails keep students informed about their library activities.

4.4 Security Techniques

To ensure system security:

- Passwords are stored using Bcrypt Hashing.
- Input validation prevents unauthorized access.
- Session-based authentication is implemented for both Admin and Student users.

5. Tools and Technologies

Component	Technology
Frontend	React.js
Styling	Tailwind CSS, Bootstrap
Backend	Node.js with Express.js

Database	MongoDB
Email Notification	NodeMailer (SMTP)
Version Control	GitHub
UI Design	Wireframes & Figma
Diagrams	Visual Paradigm
Development Tools	Visual Studio Code

6. Novelty & Innovation

The proposed Smart Library Management System (BiblioTrack) introduces several modern features that improve the efficiency and functionality of traditional library systems.

The key novelties of this project include:

- **Hybrid Library System** – The system manages both physical books and digital resources (PDFs) in a single platform, allowing students to access learning materials online as well as from the physical library.

- **Student and Admin Portals** – Separate portals are designed for students and administrators to ensure secure access and efficient management of library operations.

- **AI ChatBot Assistant** – An AI-based ChatBot is integrated to assist students with queries such as book availability, recommendations, issued books, and fine information.

- **Smart Search System** – Students can quickly search for books using different filters such as title, author, category, and availability**,** making the search process faster and more efficient.

- **Reservation System** – Students can reserve books that are currently unavailable. The system follows a first-come, first-served queue mechanism and automatically issues the book when it becomes available.

- **Fine Automation System** – The system automatically calculates fines for overdue books based on the number of delayed days.

- **E-Challan Payment System** – Overdue fines are managed through **an** E-Challan system**,** allowing students to pay fines through the 1-Bill payment method, which can later be verified by the administrator.

- **Automatic Email Notification System** – The system sends automated emails to students for important events such as account creation, book issue, reservation confirmation, and fine alerts**.**

- **Student Book Request Feature** – Students can request new books that are not currently available in the library, allowing the administration to update and expand the library collection based on student needs.

- **Clearance Report Generation** – The system allows generation of a clearance report**,** which can be used during degree clearance to verify that a student has no pending books or fines.

7. Expected Outcome

The expected outcome of this project is a fully functional web-based Smart Library Management System that will improve the efficiency of the Computer Science Department Library. The system will provide:

- Admin dashboard and student portal
- Physical book management system
- Digital library with PDF access
- Smart search functionality
- Reservation system for unavailable books
- AI ChatBot assistant for student queries
- Automated fine calculation and E-Challan system
- Email notification system for important activities
- Secure and scalable web application